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Eco-Environmental Approach to Evaluating and Optimizing Creativity and Entrepreneurship Education With PSO

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ABSTRACT

This study proposes an evaluation framework for sustainable agriculture innovation education, integrating educational inputs, rural ecological constraints, and green transformation goals. Leveraging the Particle Swarm Optimization (PSO) algorithm, it optimizes resource allocation for sustainability outcomes. PSO outperforms the SVM algorithm, improving precision agriculture resource efficiency by 4.4%, regional low-carbon farming training allocation by 9.2%, and student-driven greenhouse gas reduction by 7.8%. The framework incorporates three dimensions: agricultural platform support (e.g., university-cooperative incubators), sustainable resource allocation (e.g., precision agriculture lab funding), and sustainability outcomes (e.g., ecological literacy, farmer adoption). Using data from six agricultural universities and two rural innovation hubs in China, this evaluation system quantifies the impact of agricultural education on rural green development, contributing to global food security and low-carbon agriculture.

KEYWORDS

Sustainable Agriculture Innovation, Rural Innovation Ecosystem, PSO Algorithm, Evaluation System, Ecological Literacy

INTRODUCTION

Amid global food security challenges and the urgent need for agricultural green transformation, developing college students' sustainable agriculture innovation capabilities—such as advancing precision agriculture technologies, designing circular farming models, and leading low-carbon agribusinesses—has become a central goal of agricultural higher education (Ramos, 2024; Tiwari, 2022). As the global economy shifts toward innovation-driven development, nations increasingly depend on talents who can integrate entrepreneurial skills with ecological responsibility, particularly in agriculture, where resource constraints and environmental pressures are pronounced (Koshariya et al., 2023; Ninh, 2021). In China, this demand is intensified by national strategies such as rural revitalization and the "Dual Carbon" goal for agriculture, which call for a workforce capable of transforming green concepts into practical on-farm applications (Qin et al., 2022; Xie & Huang, 2021).

However, implementing sustainable agriculture innovation education in institutions presents distinct challenges. First, most programs remain disconnected from rural innovation systems; they emphasize classroom instruction of general business concepts with limited collaboration with

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agricultural extension centers, farmer cooperatives, or agritech startups, leaving students ill-equipped to solve real-world agricultural problems (Dai et al., 2024; Soam et al., 2023). Second, resource allocation for agricultural education is inefficient, as essential assets such as precision agriculture labs, field training plots, and agri-innovation mentorship are often concentrated in generic entrepreneurship courses, disregarding regional agricultural characteristics—for instance, omitting water-saving training for arid northern farms or organic farming guidance for southern cash crop regions (Zhang & Drury, 2024). Third, effective evaluation systems tailored to sustainable agriculture are lacking; current methods prioritize surface-level entrepreneurial outcomes, such as the number of registered agritech projects, while neglecting ecological indicators like farm resource efficiency or students' ecological literacy, creating a "green gap" in talent development (Khan et al., 2022; Li et al., 2021).

The particle swarm optimization (PSO) algorithm, a robust computational intelligence tool for solving complex optimization problems, provides unique value for addressing these challenges (Sriram & Kalyanaraman, 2025). In sustainable agriculture education, PSO can dynamically optimize the allocation of agriculture-specific resources—for instance, reallocating funds from general business workshops to precision agriculture equipment training based on regional farm needs—and integrate real-time data from agricultural Internet of Things (IoT) platforms (e.g., soil moisture sensors, greenhouse gas [GHG] monitors) to connect educational inputs with ecological outcomes (Mirzaei et al., 2024; Xu et al., 2023). This capability positions PSO as a suitable mechanism to bridge education, entrepreneurship, and rural green development.

This study seeks to address existing research gaps by constructing a sustainable agriculture innovation evaluation system that integrates the rural innovation ecosystem and is optimized using the PSO algorithm. Its innovations are twofold: first, from sociological and agricultural perspectives, it embeds sustainable agriculture education within rural innovation systems, linking agricultural universities, rural agritech enterprises, farmer cooperatives, and agricultural extension centers to ensure that education meets tangible rural ecological and economic needs; second, it replaces the metaphorical "educational eco-environment" with a practical assessment framework combining qualitative methods (e.g., interviews with agricultural extension officers) and quantitative indicators (e.g., student-driven farm water-use efficiency gains), thereby providing a comprehensive evaluation of both entrepreneurial and ecological outcomes.

RELATED WORK

Sustainable Agriculture Innovation Evaluation Systems

Research on agricultural entrepreneurship and innovation education has expanded in recent years, yet critical gaps remain in its alignment with sustainability goals and rural practical needs—issues underscored by interdisciplinary studies on agricultural education design, youth training, and evaluation frameworks. Secinaro et al. (2022), in their examination of agricultural entrepreneurship and emerging technologies, identified a pronounced homogenization problem in agricultural university curricula based on interviews with academics and industry experts. Despite distinct regional agricultural contexts—such as smallholder grain farming in sub-Saharan Africa or high-value cash crop cultivation in Southeast Asia—courses seldom adapt to local farm needs. For example, curricula emphasizing generic precision agriculture technologies (e.g., drone mapping) often exclude training in context-specific tools like low-cost soil moisture sensors for resource-limited smallholders or urban farming techniques for peri-urban settings. This uniform approach produces graduates with technical knowledge but limited capacity to tackle region-specific ecological challenges (e.g., soil erosion in hilly terrain or water scarcity in arid zones), diminishing the educational program's practical and ecological relevance.

Adeyanju et al. (2021) evaluated the influence of agricultural training programs on youth agricultural entrepreneurship performance in Kenya, highlighting that while these programs enhance business management competencies (e.g., crop marketing, cost control), they largely neglect the development of ecological responsibility. Their findings revealed that only 12% of trained youth adopted sustainable practices (e.g., reduced chemical use, crop rotation) in their agribusinesses, primarily because the training emphasized individual entrepreneurial outcomes rather than educational systems fostering sustainability. For instance, the programs lacked field-based modules that pair students with organic farmers to learn soil conservation or agri-waste recycling, failing to establish a structural framework for institutions to cultivate green entrepreneurs who balance profitability with environmental stewardship.

Pan et al. (2024), in their study on the influence of farmer entrepreneurship on rural economic growth from an innovation-driven perspective, proposed a support framework for agricultural innovation integrating government policy incentives, agritech enterprise partnerships, and university technical mentorship. Although this model acknowledged the importance of external support in promoting agricultural entrepreneurship, it explicitly omitted agricultural extension centers—key actors in converting student and farmer innovations into practical field applications. For example, student-developed drip-irrigation systems or organic fertilizer formulations often fail to reach smallholder farmers because no extension officers are available to demonstrate usability or adapt technologies to local conditions. This omission restricts the framework's capacity to generate measurable ecological and economic benefits in rural areas, as innovations remain confined to academic or enterprise contexts instead of being implemented by the farmers who need them most.

Zhou et al. (2022) developed a quality evaluation system for college students' agricultural innovation and entrepreneurship education using extenics, a methodology for resolving contradictory problems. Their framework incorporated indicators such as "business plan feasibility," "project registration rate," and "student startup survival rate," but notably omitted metrics related to ecological sustainability. For instance, the system did not evaluate students' capacity to conduct life-cycle assessments for agricultural products (e.g., calculating the carbon footprint of vegetable cultivation) or design projects that reduce agricultural waste (e.g., converting crop straw into livestock feed). Consequently, the evaluation failed to capture the multifaceted nature of sustainable agricultural innovation, which requires balancing economic feasibility with environmental stewardship, leaving institutions unable to determine whether their programs cultivate entrepreneurs who advance rural green transformation.

Arief et al. (2021) further addressed these shortcomings by examining online entrepreneurship education programs designed to train farmer entrepreneurs through urban farming. Their study revealed that although online platforms expanded access to education for rural and peri-urban youth, the programs remained disconnected from local ecological contexts. For example, courses on urban hydroponics seldom adapted content to regional water conditions (e.g., simplifying systems for water-scarce regions) or local crop preferences (e.g., emphasizing leafy greens in temperate zones while neglecting tropical vegetables). This inconsistency underscores a persistent challenge in agricultural education: even technologically advanced delivery models fail to align with regional ecological needs, limiting their potential to promote sustainable, context-sensitive entrepreneurship.

Rural Innovation Ecosystem Concept

The rural innovation ecosystem concept, when applied to agricultural education, highlights the interdependent relationships among agricultural institutions and rural ecological, economic, and social systems—an interdependence increasingly supported by recent research on regional innovation dynamics and multi-actor collaboration. Bankova and Tsvetanova (2023), in their study on rural development through regional innovation ecosystems (RIEs), defined the agricultural education ecosystem as a subset of broader RIEs, where educational institutions (e.g., agricultural universities) collaborate with rural actors (e.g., agritech enterprises, farmer cooperatives) to balance three core dimensions: student skill development, rural resource utilization (e.g., local soil, water, and crop varieties), and sustainable economic outcomes. They emphasized that effective agricultural education must embed classroom learning within on-farm problem-solving—for instance, partnering with local farms to use real-time soil moisture data collected through rural IoT networks to teach students data-driven nutrient management—rather than relying on abstract, uniform theories. While their research identified RIEs as a framework for aligning education with rural needs, it also noted that most agricultural programs lack concrete tools to operationalize this balance (e.g., metrics evaluating how well curricula respond to local farm challenges).

Bravaglieri et al. (2025) expanded this perspective by examining multi-actor rural innovation ecosystems, stressing that sustainable agriculture innovation education cannot be separated from the dynamic interactions among rural stakeholders. Their analysis of European rural regions demonstrated that educational content must correspond to both local farming practices and ecological limitations—for example, in maize-dominated regions of Eastern Europe, agri-waste recycling training should focus on converting maize straw into mushroom substrate, a locally viable technique, rather than on generic composting. However, their study also identified a critical gap: although multi-actor collaboration among universities, extension centers, and farmers is vital for contextualizing education, few studies have proposed practical methods for embedding rural eco-environmental characteristics into educational evaluation systems—such as developing rubrics to assess whether curricula address region-specific ecological risks (e.g., soil erosion in hilly regions or salinity in coastal farms).

Hammer and Frimanslund (2022), drawing on rural ecosystems in Norway, introduced the concept of “contextualized innovation harmony” to describe the alignment between educational outputs (e.g., student-designed agritech projects) and the ecological carrying capacity of rural environments. For instance, in arid rural regions, their research showed that effective agricultural education programs emphasize water-saving technology training (e.g., low-pressure drip irrigation) rather than water-intensive projects (e.g., traditional hydroponics), which would worsen local water scarcity. This study offered a theoretical basis for understanding how education can operate within rural ecological constraints but did not address how to optimize resource distribution across educational components (e.g., balancing funding for water-saving laboratories versus farmer extension workshops) to sustain such harmony.

Thomas et al. (2021) examined the role of universities as “orchestrators” of regional innovation ecosystems in emerging economies, demonstrating that systemic thinking—central to ecosystem theory—can enhance the effectiveness of agricultural education. Their case studies of African agricultural universities revealed that institutions actively collaborating with local agritech startups and government extension services are better equipped to design curricula that promote both innovation and sustainability. For example, a Kenyan university partnered with local mobile technology firms to develop a course on “digital market linkages for smallholder farmers,” which not only strengthened students’ entrepreneurial skills but also increased farmers’ crop sales by 30%. While this work highlights the potential of ecosystem theory to inform agricultural education, it also noted that integrating computational intelligence tools (e.g., PSO algorithms for resource optimization) into such orchestration remains underexplored, leaving a gap in quantifying and refining educational resource allocation across multi-actor networks.

Yin et al. (2022), in their study of rural innovation systems for sustainable revitalization, expanded these insights by linking ecosystem theory to measurable rural development outcomes. They argued that agricultural education evaluation should include indicators reflecting how effectively programs contribute to rural innovation ecosystem functions—such as “student participation in farmer-led innovation projects” or “adoption rate of student-developed technologies by local cooperatives.” This approach advances beyond theoretical discussions of ecosystem alignment by proposing practical evaluation benchmarks, though it still lacks strategies for integrating these indicators with algorithmic tools (e.g., PSO) to optimize resource allocation for both educational and ecological objectives.

Overall, previous research has significantly advanced understanding of sustainable agriculture education and rural innovation ecosystems; however, three key gaps remain. First, few studies integrate agricultural education within rural innovation systems, resulting in a persistent disconnect between education and practice. Second, existing evaluation frameworks seldom include ecological metrics, limiting their ability to assess the sustainability of agricultural innovation. Third, no research has yet combined the rural innovation ecosystem concept with the PSO algorithm to optimize resource allocation for sustainable agriculture education. This study addresses these gaps by developing a context-specific, PSO-optimized evaluation system that links agricultural education with rural green transformation.

MATERIALS AND METHODS

Sample and Data Collection

Sample Selection

To ensure alignment with rural innovation systems and representativeness across diverse agricultural contexts, the study selected eight institutions across two Chinese provinces with distinct agricultural characteristics: five agricultural colleges and vocational technical schools in Henan Province—including Henan Agricultural Vocational College and Xinxiang Vocational and Technical College, both emphasizing grain production education—and three agricultural universities in Jiangsu Province, including Nanjing Agricultural University and Yangzhou University, which specialize in high-value cash crop and agritech innovation. The sample comprised 1,200 undergraduate students from the 2023–2024 academic year, all majoring in disciplines directly related to sustainable agriculture: Agricultural Resources and Environment (400 students, 33.3%, focused on soil and water resource management), Agronomy (500 students, 41.7%, specializing in low-carbon crop cultivation), and Agribusiness Management (300 students, 25.0%, focused on green agribusiness operations). This selection ensured comprehensive coverage of core sustainable agriculture competencies and alignment with regional agricultural priorities—grain production in Henan and high-tech agriculture in Jiangsu.

Data Types and Sources

Data were categorized as primary (education- and stakeholder-focused) and secondary (rural impact and institutional) to comprehensively capture the relationship between sustainable agriculture education and rural green development. Primary data were collected through structured questionnaires and semi-structured interviews to assess educational processes and stakeholder needs. The questionnaire was developed around eight sustainable agriculture–specific dimensions: ecological literacy (e.g., understanding of circular agriculture), low-carbon farming skills (e.g., proficiency in precision irrigation), agri-innovation design (e.g., ability to create agri-waste recycling solutions), rural collaboration (e.g., experience working with farmer cooperatives), sustainable resource utilization (e.g., optimization of fertilizer use), eco-entrepreneurial practice (e.g., participation in agritech startups), sustainability awareness (e.g., recognition of agricultural carbon emission risks), and regional agricultural adaptation (e.g., knowledge of local crop-specific sustainable methods). It included 35 items, with reliability and validity verified using Cronbach's $\alpha = 0.89$ and Kaiser-Meyer-Olkin = 0.84, indicating strong internal consistency and sampling adequacy.

Semi-structured interviews were conducted with 15 experts representing key actors in the rural innovation ecosystem: five agricultural university educators (with ≥ 5 years of experience in sustainable agriculture instruction), five agritech enterprise managers (from firms specializing in precision agriculture or organic fertilizer development), three local agricultural extension officers (responsible for promoting sustainable technologies to smallholders), and two farmer cooperative leaders (managing grain or cash crop operations). The interviews focused on validating the relevance of evaluation indicators (e.g., "Does 'agri-waste recycling project count' reflect real rural needs?") and identifying regional educational priorities (e.g., water-saving training for Henan's dryland farming).

Secondary data were obtained from two sources to quantify institutional resources and rural impacts. Institutional records captured agricultural education–specific inputs, including the number of full-time faculty specializing in sustainable agriculture, annual funding for precision agriculture labs (e.g., IoT soil sensor procurement), area of on-campus agri-incubators, and number of field training plots partnered with local farms. Rural impact data were collected from local agricultural bureaus and cooperative farms, covering adoption rates of student-developed sustainable technologies (e.g., organic pest control methods), GHG reduction rates in farms collaborating with student teams (measured via on-farm IoT monitors), and survival rates of student-founded agritech startups focused on sustainability. All secondary data were cross-verified with official reports to ensure accuracy.

Analytical Tools

The following four primary tools were used to process data, optimize resources, and validate the evaluation system, supplemented by agricultural-specific data collection platforms:

- MATLAB R2022b: Implemented the PSO algorithm customized for sustainable agriculture education, including parameter calibration (e.g., setting inertia weight ranges for regional resource allocation), simulation of resource optimization scenarios (e.g., balancing funding between precision agriculture laboratories and low-carbon training programs), and convergence testing to ensure algorithm stability in agricultural contexts (e.g., adapting to seasonal variations in farm training demands).
- Statistical Package for the Social Sciences (version 26.0): Preprocessed questionnaire data (e.g., min–max normalization of ecological literacy scores, factor analysis to refine sustainable agriculture skill dimensions) and performed descriptive statistics (e.g., calculating the percentage of students proficient in precision irrigation operations, analyzing regional differences in agri-innovation participation).
- AMOS (version 24.0): Validated the structure of the sustainable agriculture innovation evaluation system through confirmatory factor analysis, testing whether the three hierarchical levels—platform support, resource allocation, and sustainability outcomes—and 28 indicators constituted a coherent framework.
- Agricultural IoT Platforms: Collaborated with local agritech firms using IoT-based monitoring systems (e.g., soil moisture sensors, portable GHG detectors) to collect real-time on-farm data. These data were integrated into the PSO algorithm's fitness function to link educational resource inputs with measurable ecological outcomes (e.g., student training hours relative to farm water-use efficiency gains).

Eco-Environment Optimization of Entrepreneurship and Creativity With PSO

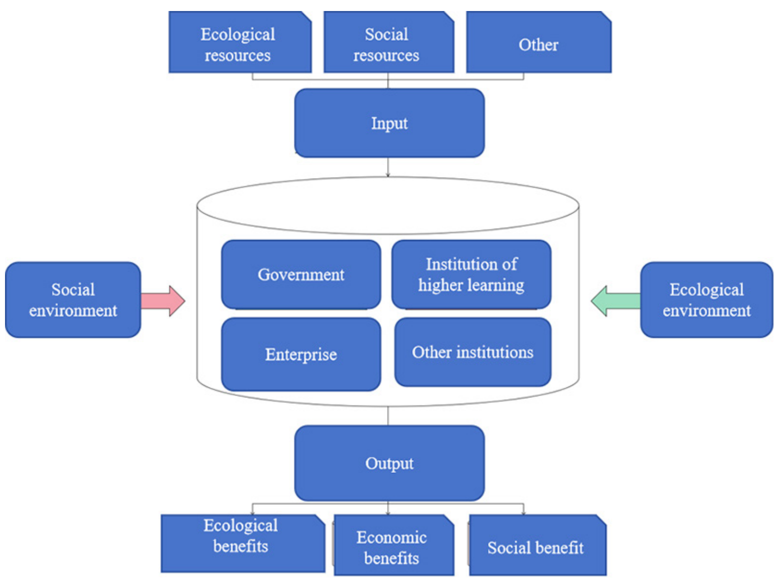
Sustainable agriculture innovation education in agricultural institutions is a complex systemic process encompassing the sustainable agriculture innovation ecosystem, which integrates agricultural education inputs (e.g., precision agriculture laboratories, on-farm training resources, eco-agritech mentorship), rural ecological constraints (e.g., regional water scarcity, soil fertility levels), and tangible outputs (e.g., student-developed low-carbon farming solutions, farmer-adopted agritech tools) (Athuman, 2023; Moschitz & Home, 2014). To improve the efficiency and sustainability of this system, the PSO algorithm can be tailored to address agriculture-specific resource allocation and ecological alignment challenges.

The first step in this optimization process is refining the agricultural education material environment and balancing the rural–education ecology. From the perspective of agricultural educational ecology, the core of sustainable agriculture innovation education lies in aligning educational resources with both student skill development and rural ecological needs. Here, "ecology" denotes the effective matching of educational inputs to on-farm practical demands—for instance, prioritizing precision agriculture equipment (e.g., IoT soil moisture sensors, drone mapping tools) for students in arid regions such as Henan's dryland farming zones or allocating organic fertilizer research and development laboratory resources to Jiangsu's cash crop-producing counties, where chemical fertilizer reduction is a key policy goal. This balance ensures that educational resources are not isolated but instead directly support the resolution of local sustainability challenges such as soil degradation and water waste.

The PSO algorithm plays a pivotal role in this alignment by dynamically adjusting the distribution of agricultural education resources. Unlike generic optimization models, PSO integrates real-time rural ecological data (e.g., farm GHG emission levels, crop water demand) and student skill profiles to iteratively identify optimal resource configurations. For example, PSO can determine the ideal allocation of annual agricultural education funding between two priorities: precision irrigation training programs, vital in water-scarce regions, and agri-waste recycling workshops, essential in areas with high crop residue output. It can also optimize faculty time distribution—defining how many hours instructors devote to classroom teaching (e.g., explaining circular agriculture principles) versus on-farm supervision (e.g., assisting students in testing straw composting techniques with local cooperatives)—to maximize both student competence and on-farm impact (Cui, 2023; Moazeni et al., 2023).

As illustrated in Figure 1, the system comprises government agencies, higher-learning institutions, enterprises, and other social actors that interact within the agricultural education ecosystem. The PSO algorithm can optimize the relationships and resource flows among these entities—for instance, identifying the ideal level of collaboration between universities and enterprises to ensure students receive industry-relevant, hands-on training while allowing firms to benefit from innovative ideas and emerging talent. In this way, the overall educational ecology becomes more balanced, and the material environment is optimized to better support entrepreneurship and creativity in sustainable agriculture.

Figure 1. Primary Model of Ecological System of Entrepreneurship and Creativity



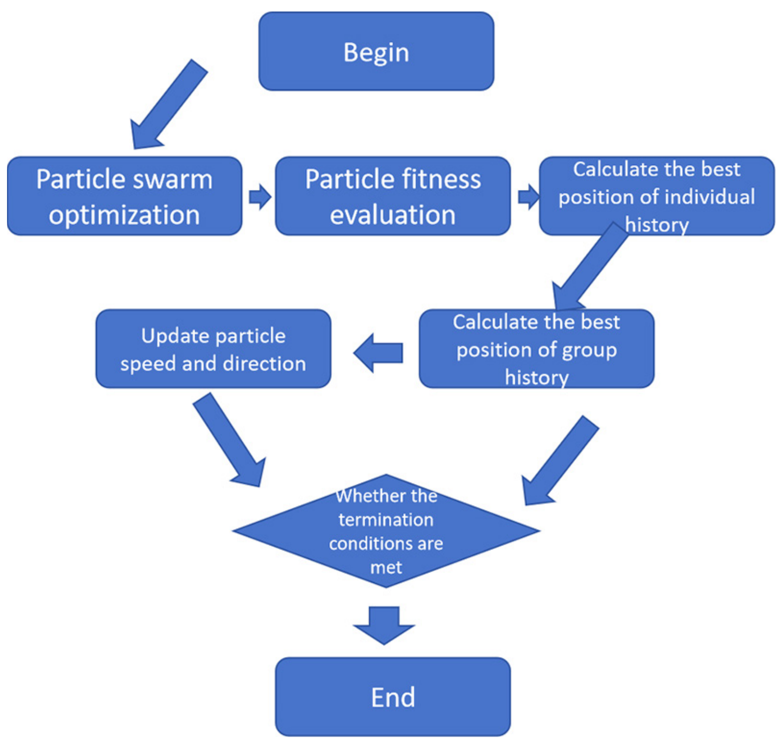
The eco-environment for training entrepreneurship and creativity talent functions as a dynamic system designed to meet the information needs of its participants. This is achieved through the continuous absorption and filtration of relevant information, guided by tutors within higher education contexts. Within this environment, multiple ecological factors interact and depend on one another. This interaction mirrors the evolutionary process in genetic algorithms, wherein the PSO algorithm generates an initial set of solutions and iteratively searches for training errors. These errors are then used to calculate the fitness value using Equation 1:

$$f(x) = \frac{1}{1 + \frac{1}{2n}\sum_{p=1}^{n}(y_p - t_p)} \tag{1}$$

Where p is the Number of samples, y_p represent the actual output value and t_p is the output value in the sample.

At the start of the algorithm's iteration, larger inertia weight values are selected to strengthen the algorithm's global optimization capability. As the iteration progresses, smaller inertia weights are applied to enhance its capacity for local optimization. The flow of the basic PSO algorithm is illustrated in Figure 2 below.

Figure 2. PSO Algorithm Flow



The second aspect involves analyzing the geographical environment to shape educational characteristics. Educational activities represent essential dimensions of human development, defining the causal relationship between the eco-environment of these activities and the geographical environment in determining the quality of entrepreneurship and creativity talent training. In other words, the quality of entrepreneurship and creativity talent development differs across information environments—some align with societal needs, while others exhibit significant disparities. The allocation problem is a complex, nonlinear optimization issue aimed at maximizing educational outcomes. Accordingly, the efficiency of entrepreneurship and creativity education resource utilization is determined by the ratio of educational output to educational input and can be expressed using Equation 2:

$$U = \frac{\sum_{j=1}^{m} \mu_i O_i}{\sum_{i=1}^{n} \phi_i I_i} \tag{2}$$

Where:
i—Elements of educational resources
j—Elements of educational achievements;
O_i I_i—Output and input of educational resources
ϕ_i μ_i—Weight of each educational resource input index and output index
To evaluate the effectiveness of goals and requirements for constructing the entrepreneurship and creativity education eco-environment, it is essential to develop an evaluation system for the school's entrepreneurship and creativity education eco-environment that meets updated standards. This system functions as a key instrument for advancing the ongoing reform of entrepreneurship and creativity education in schools. Therefore, it is critical to maintain high diversity within the search space to increase the likelihood of identifying the global optimal solution, while simultaneously reducing the negative effects of excessive diversity on algorithmic convergence. Excessive diversity may decrease solution accuracy or even hinder convergence. Consequently, improving and refining population-based intelligent optimization algorithms is essential. Population diversity strongly influences the algorithm's global convergence performance and can be represented by Equation 3:

$$diversity(S) = \frac{1}{|S| * |L|} * \sum_{i=1}^{|S|} \sqrt{\sum_{i=1}^{N} (p_i - p_g)^2} \tag{3}$$

Where:
s—Population;
|S|—Number of particles in the population;
|L|—The longest radius of the search space;
N—Dimension of the problem
Using random distribution to determine inertia weights and the time-varying acceleration factor enhances the algorithm's global search capability. However, the randomness of inertia weights introduces significant variability in the number of iterations required for convergence, resulting in inconsistent performance improvements (Fatimah & Jayashree, 2026). To address this issue, the total population within the algorithm is divided into several subpopulations. In each generation, particle positions change according to time-varying inertia weights and acceleration factors, while selected particles from all subpopulations are disrupted and recombined to form new subpopulations. This method increases population diversity. The multi-objective optimization function for entrepreneurship and creativity resource allocation can then be expressed using Equations 4 and 5:

$$\max U_k = \frac{\sum_{j=1}^{m} \mu_i O_i}{\sum_{i=1}^{n} \phi_i I_i} \tag{4}$$

$$\max F_k = \sum_{i=1}^{n} \phi A_{ki}, k = 1, 2, ..., k \tag{5}$$

Where:

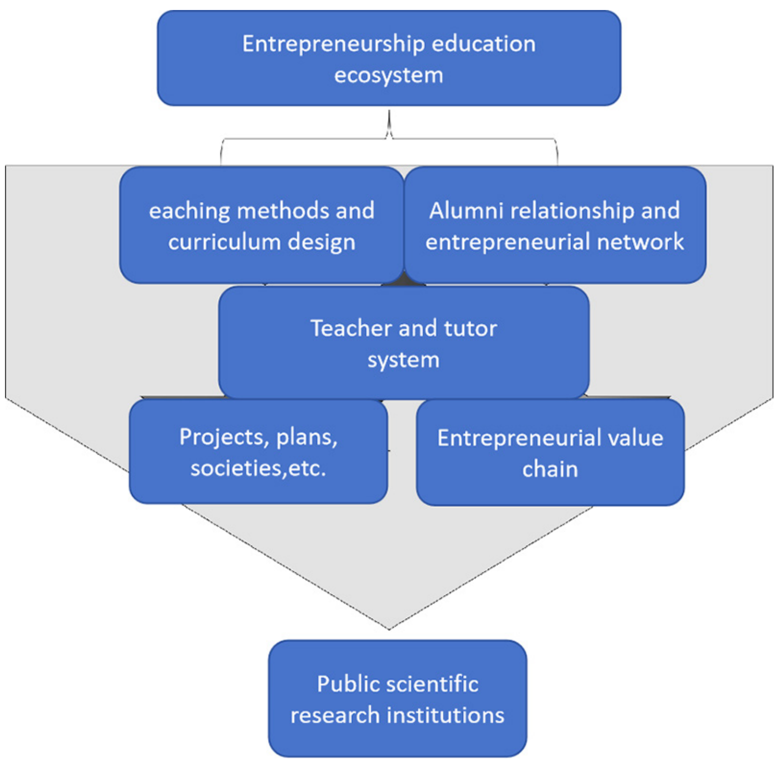
max U_k—The utilization efficiency of educational resources in each school should seek the maximum value as far as possible.

max F_k—The allocation efficiency of educational resources in each school should seek the maximum value as far as possible.

Construction of the Entrepreneurship and Creativity Evaluation System

In evaluating entrepreneurship and creativity education, excessive emphasis is often placed on visible outcomes such as competition rankings, project submission counts, and participant numbers (Mehta, 2025; Sriram & Kalyanaraman, 2025). However, diagnostic, formative, and summative assessments of entrepreneurship and creativity education in schools are frequently neglected, resulting in incomplete and inaccurate evaluation systems. Developing an assessment framework for entrepreneurship and creativity education based on the eco-environment concept requires a clear definition of its core constructs, along with explicit articulation of functions, principles, and categories within school-based entrepreneurship and creativity education. This framework also provides theoretical guidance for future research. The entrepreneurship network comprises four institutional types: college-level, interdisciplinary, technology transfer, and academic organizations. The university entrepreneurship education ecosystem is illustrated in Figure 3.

Figure 3. Schematic Diagram of University Entrepreneurship Education Ecosystem



First, the assessment indicators for entrepreneurship and creativity were developed through an in-depth analysis of existing studies. The currently recognized indicators were summarized, and a cluster of indicators consistent with the assessment of college students' entrepreneurship and creativity was constructed through a comprehensive literature review. Consequently, the overall index value is not a simple sum of all sub-indicators but a weighted summation relationship, as expressed in Equation 6:

Where:

$$S = \sum_{i=1}^{n} w_i f_i(I_i), i = 1, 2, ..., n \tag{6}$$

$f_i(I_i)$—Some measure of I_i

w_i—Weight value of each index

The school entrepreneurship and creativity education eco-environment can be described across three levels: the platform support environment, the resource allocation environment, and the educational outcomes of entrepreneurship and creativity programs. These levels form the foundation of the evaluation index system for entrepreneurship and creativity education environments in schools. To integrate the development of the geographical environment with entrepreneurship and creativity education as a unified system, the educational content should not focus solely on the internal advancement of entrepreneurship and creativity education. It must also reflect the needs of local economic and cultural development, transitioning from a narrow disciplinary focus to a more comprehensive approach. Accordingly, the content of entrepreneurship and creativity education should align with regional economic and cultural contexts, evolving from a single-focus curriculum to a more holistic educational model. In this context, larger inertia weights enhance the algorithm's global search capability, while smaller inertia weights strengthen its local search performance. The inertia weight variation equation for the linear decreasing weight strategy is therefore expressed using Equation 7:

$$\omega(t) = \omega_{ini} + \frac{\omega_{\delta i} - \omega_{ini}}{t_{\max}} \cdot t \tag{7}$$

Where:

$\omega(t)$—Value of inertia weight

t—Number of current iterations

t_{max}—Maximum number of iterations

ω_{int}—Initial inertia weight of

$\omega_{\delta i}$—Final inertia weight

Second, the indicator clusters were analyzed through a review of relevant literature, followed by expert interviews and analysis of questionnaire results. Indicators deemed inapplicable were removed, while essential ones were added to establish a "pool of alternative indicators for measuring the entrepreneurship and creativity of college students." The input sample normalization is expressed using Equation 8:

$$X = \frac{T - T_{\min}}{T_{\max} - T_{\min}} \tag{8}$$

Where:

T_{min}—Enter minimum value.

T_{max}—Enter the maximum value;

T—Unprocessed input value;

The entrepreneurship and creativity platform support environment provides essential support for students' entrepreneurial activities within schools and represents a core component of the school's entrepreneurship and creativity education environment evaluation system. Through the analysis of random variables, a Gaussian distribution hypothesis is proposed to replace the uniform distribution. The changes are expressed using Equation 9:

$$x_{id}{}^{t} = G(p_{id} + g_{id})\frac{|p_{id} - g_{id}}{2} \tag{9}$$

Where:
G——Normal distribution
p_{id}, g_{id}—The center point of oscillation
The regional characteristics of entrepreneurship and creativity education are reflected not only in the administrative connections between schools and their local areas but also in the educational development process itself. A strong linkage between the two is essential to ensure that the goals of entrepreneurship and creativity education are shaped by the specific geographical and industrial needs of the region. Leveraging local geographical and environmental advantages can promote the distinctive development of entrepreneurship and creativity education.

The pool of alternative elements for evaluation indicators, reviewed by experts, constitutes the foundational upper-level indicators grounded in theoretical principles. The weight of each indicator is determined using expert scoring tables, resulting in the formation of the initial assessment index system. Through empirical analysis based on interaction data, the final "College Students' Entrepreneurship and Creativity Ability Assessment Index System" is established. Subsequently, the correlation coefficient matrix of the evaluation indicators is calculated.

$$R = (r_{ij})_{p\times p} \tag{10}$$

Where: R_{ij}—Correlation coefficient between i teaching quality evaluation sample and j index

RESULTS AND ANALYSIS

Analysis of Comprehensive Learning PSO Algorithm With Evaluation Mechanism

The convergence and performance of the PSO algorithm in sustainable agriculture innovation education are critically affected by parameters such as acceleration factors and maximum velocity, which must be calibrated to address agriculture-specific resource allocation challenges. Poor parameter settings may prevent the algorithm from balancing key priorities—for instance, overinvesting in precision agriculture laboratories while neglecting low-carbon farming training or misallocating resources across regional agricultural contexts (e.g., emphasizing water-saving technologies in humid areas). Such misalignment can undermine the algorithm's capacity to achieve tangible ecological outcomes, including reductions in farm GHG emissions and improvements in water-use efficiency.

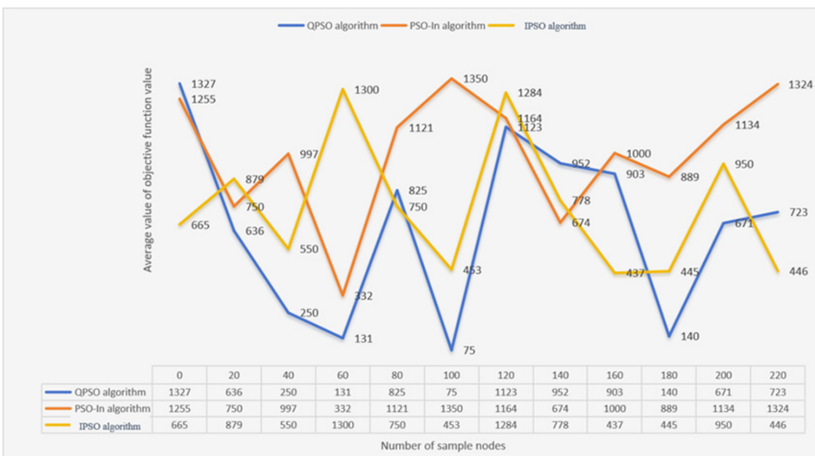
To overcome these challenges, improvements to the PSO algorithm focused on enhancing convergence speed, avoiding local optima (e.g., prioritizing short-term entrepreneurial outcomes over long-term ecological impact), and maintaining population diversity to accommodate variable rural needs (e.g., seasonal shifts in training priorities between planting and harvest periods). Enhancements included adjusting particle neighbor relationships to integrate real-time data from agricultural extension centers (e.g., farmer feedback on unmet training needs) and refining particle topology to represent the interconnected components of rural innovation systems, such as universities, agritech startups, and cooperatives.

Within sustainable agriculture education, PSO optimizes the allocation of agricultural-specific resources—such as precision agriculture laboratory equipment, on-farm internship slots, and agri-waste recycling workshop funding—to maximize three interrelated objectives: student mastery of sustainable skills, adoption of student-developed technologies by farmers, and measurable ecological improvements on partner farms. This process requires balancing trade-offs, such as allocating funds between soil sensor training (to reduce fertilizer use) and drone irrigation management (to conserve water), according to regional agricultural priorities (e.g., water scarcity in Henan versus soil health in Jiangsu).

To verify PSO's performance, comparative experiments were conducted using three variants—Quantum-behaved Particle Swarm Optimization (QPSO), Particle Swarm Optimization with Inertia weight (PSO-In), and Improved Particle Swarm Optimization (IPSO)—across 100 simulations. Each simulation optimized the distribution of two critical resources: precision agriculture laboratory funding and low-carbon farming training hours. The objective function in each simulation integrated three agricultural sustainability metrics: (a) student proficiency in sustainable practices (e.g., precision irrigation operation), (b) farmer adoption rates of student-developed technologies, and (c) GHG reduction on partner farms, measured through IoT-based sensors.

As depicted in Figure 4, the basic PSO algorithm outperformed its variants in balancing these objectives. Its average objective function value stabilized at 75 by iteration 100, demonstrating consistent optimization of resource allocation to enhance both ecological and educational outcomes. In contrast, QPSO displayed greater volatility, fluctuating between 131 and 90 in later iterations, while PSO-In exhibited severe instability, peaking at 1,350 by iteration 100, indicating its inability to adapt to the variability of rural agricultural needs. These results confirm PSO's suitability for sustainable agriculture contexts, where resource allocation must dynamically respond to student learning progress and on-farm ecological feedback.

Figure 4. Average Objective Function Values from Simulation Results of Different Algorithms



Note. QPSO = Quantum-behaved Particle Swarm Optimization; PSO-In = Particle Swarm Optimization with Inertia weight; IPSO = Improved Particle Swarm Optimization.

The sustainable agriculture innovation ecosystem consists of three interconnected levels: (1) agricultural education platforms, including universities, agri-incubators, and extension centers; (2) resource allocation for sustainable training, such as precision agriculture laboratories and farmer cooperative partnerships; and (3) sustainability outcomes encompassing ecological literacy, on-farm resource efficiency, and agritech adoption. To strengthen this ecosystem, PSO incorporates a "green incentive mechanism" that prioritizes resources for educational programs demonstrating verified ecological benefits—for example, allocating additional funding to student projects that reduce farm chemical use by at least 20%, while deprioritizing those with minimal environmental impact.

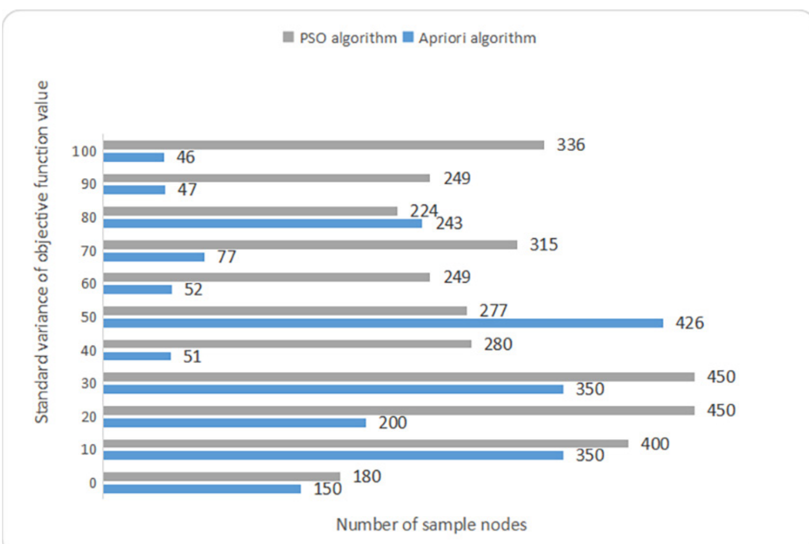
Simulation results for the basic PSO algorithm (Table 1) confirm its stability in optimizing agricultural resources. For the three core functions—F1 (precision agriculture laboratory utilization), F2 (low-carbon training effectiveness), and F3 (farmer adoption of student-developed technologies)—the average global minimum values (0.09834, 0.17264, and 0.24684) and low standard deviations (≤0.17352) demonstrate consistent performance. This stability is crucial in agricultural contexts, where erratic resource allocation could disrupt seasonal training cycles and hinder sustainable outcomes.

Table 1. Simulation Results of Basic PSO Algorithm

Function	F1	F2	F3
Global minimum point	(-0.99876,0.15675)	(-0.33415,0.19876)	(-0.11876,0.23145)
Average value of global minimum	0.09834	0.17264	0.24684
Standard deviation of global minimum	0.08425	0.09285	0.17352

During iteration, particles representing resource allocation scenarios are evaluated using a fitness function that integrates agricultural IoT data (e.g., real-time farm water use and soil nutrient levels). Particles with lower fitness values (e.g., allocations that fail to reduce fertilizer use) are replaced by higher-performing ones (e.g., allocations achieving a 15% increase in farmer adoption of organic pest control). A comparison of standard variances between PSO and the Apriori algorithm (Figure 5) indicates that although Apriori exhibits lower volatility in early iterations, PSO's variance stabilizes by iteration 100 (46 versus Apriori's 52), demonstrating its adaptability to dynamic rural data, such as sudden changes in rainfall affecting irrigation training requirements. This stability results from parameter tuning, including dynamic inertia weight adjustment and subpopulation recombination, which balance the exploration of new resource combinations with the exploitation of validated allocations.

Figure 5. Standard Variance of Objective Function Values for Different Algorithms



Note. PSO = Particle Swarm Optimization

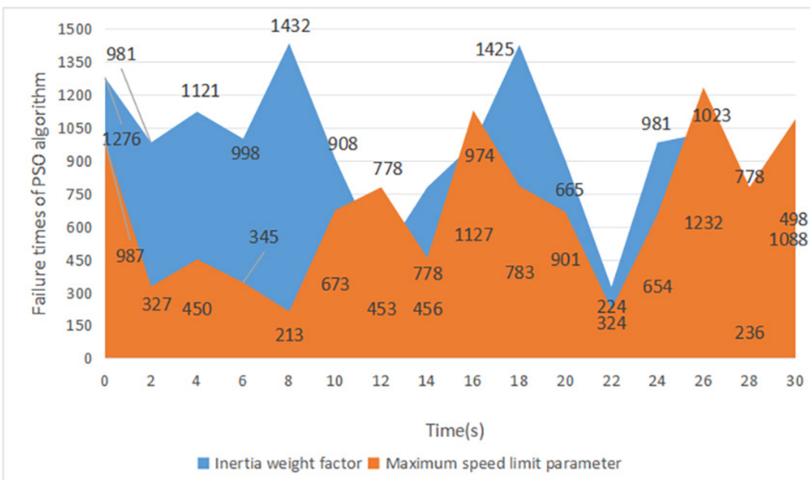
Convergence Analysis of PSO Based on Eco-Environment Concept

Indicator standardization is essential for evaluating sustainable agriculture outcomes, as metrics such as “ecological literacy scores” (0–100) and “farm GHG reduction rates” (0–50%) operate on different scales. For PSO, convergence to a stable solution—where resource allocation simultaneously maximizes educational and ecological objectives—is critical, since particles representing allocation scenarios must align with the dynamic interactions among rural innovation system components (e.g., universities updating curricula in response to farmer feedback).

Impact of Maximum Speed Limit and Inertia Weight on Algorithm Failure

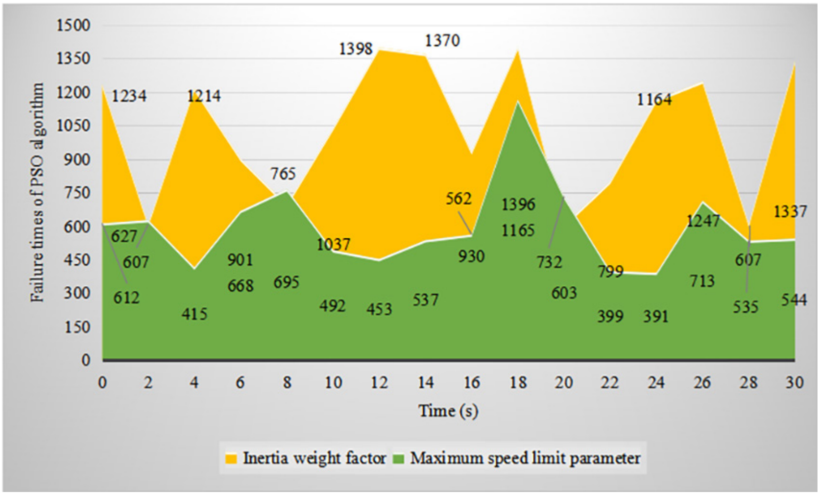
To evaluate how maximum velocity and inertia weight influence PSO’s convergence in agricultural applications, we tested failure frequency—the inability to optimize resource allocation for regional needs—under varying parameter settings. Results (Figures 6 and 7) indicate that the inertia weight factor exerts a stronger effect on stability. At an inertia weight of 10 (Figure 6), failure occurrences peaked at 1,432 (8s) and 1,425 (18s), caused by excessive allocation to generic agribusiness training while neglecting region-specific needs, such as drought-resistant crop instruction in Henan. At an inertia weight of 20 (Figure 7), failure occurrences remained high (peaking at 1,398 at 10s) but stabilized more quickly, as the higher weight reduced abrupt shifts among resource categories.

Figure 6. Failure Times of PSO Algorithm When Inertia Weight Factor Is 10



Note. PSO = Particle Swarm Optimization;

Figure 7. Failure Times of PSO Algorithm When Inertia Weight Factor Is 20



Note. PSO = Particle Swarm Optimization;

The maximum velocity parameter, set to 30% of the resource search space (e.g., 30% of total precision laboratory funding), exhibited lower volatility, with peaks at 1,023 (26s) and 1,088 (30s). These findings informed parameter tuning, where a linearly decreasing inertia weight (from 0.9 to 0.4) combined with a maximum velocity of 0.3×search space minimized failures. This configuration enabled PSO to adaptively balance short-term training priorities (e.g., pre-planting workshops) with long-term ecological objectives (e.g., annual GHG reduction targets).

Comparison of Resource Utilization and Allocation Efficiency Between PSO and Support Vector Machine Algorithms

To quantify PSO's effectiveness in enhancing sustainable agriculture education, its performance was compared with that of the support vector machine (SVM) algorithm across three agricultural-specific metrics (Table 2). PSO outperformed SVM in all categories:

- Precision agriculture resource utilization efficiency: 0.129 (PSO) vs. 0.085 (SVM), reflecting superior allocation of soil sensors and drone equipment to student projects with high on-farm impact.
- Regional resource allocation efficiency: 0.227 (PSO) vs. 0.135 (SVM), driven by PSO's capacity to prioritize water-saving training in Henan and organic farming laboratories in Jiangsu.
- Student-driven farm GHG reduction rate: 12.8% (PSO) vs. 5.0% (SVM), as PSO-directed resources emphasized low-carbon practices (e.g., crop–livestock integration) adopted by partner farms.

Table 2. School Resource Utilization Efficiency and Resource Allocation Efficiency

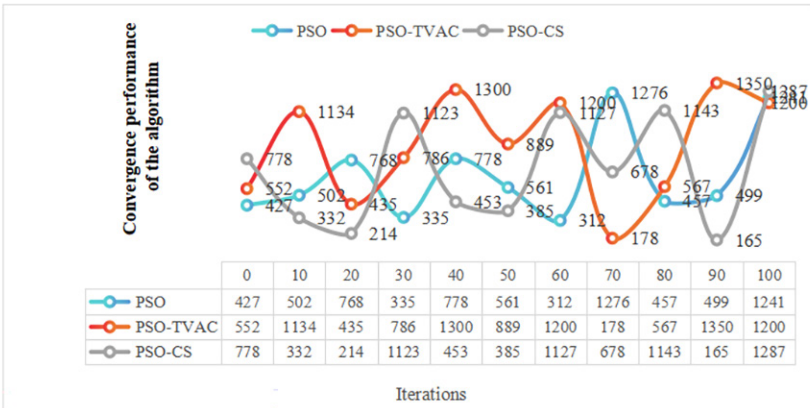
	PSO	SVM	Difference
Resource Utilization Efficiency	0.129	0.085	0.044
Resource allocation efficiency	0.227	0.135	0.092

Note.PSO = Particle Swarm Optimization;SVM = Support Vector Machine.

Convergence Performance of PSO Variants

To evaluate the effectiveness of different PSO variants, three algorithms were tested: the standard PSO, the PSO with time-varying acceleration coefficients, and the PSO with crossover strategy. Each algorithm was executed 20 times for each problem, and the experimental results are depicted in the Figure 8 (horizontal axis: iterations; vertical axis: algorithm convergence performance).

Figure 8. Convergence Performance of Different Algorithms



	0	10	20	30	40	50	60	70	80	90	100
PSO	427	502	768	335	778	561	312	1276	457	499	1241
PSO-TVAC	552	1134	435	786	1300	889	1200	178	567	1350	1200
PSO-CS	778	332	214	1123	453	385	1127	678	1143	165	1287

Note.PSO = Particle Swarm Optimization; PSO-TAVC=the PSO with time-varying acceleration coefficients;PSO-CS=the PSO with crossover strategy.

PSO (Turquoise Line). The convergence process is relatively stable. At iteration 0, the performance value is 427; it increases to 768 at iteration 20, then fluctuates (e.g., dropping to 335 at iteration 30 and 312 at iteration 60) before peaking at 1,276 at iteration 70 and 1,241 at iteration 100. These results demonstrate that PSO maintains steady optimization capability, avoiding extreme fluctuations while achieving high convergence performance in later iterations.

PSO With Time-Varying Acceleration Coefficients (Red Line). This variant shows rapid early-stage convergence but suffers from severe volatility. At iteration 40, it peaks at 1,300, drops sharply to 178 at iteration 70, and then spikes again to 1,350 at iteration 90. Such drastic fluctuations indicate poor stability, making the algorithm susceptible to local optima or convergence stagnation.

PSO With Crossover Strategy (Gray Line). The convergence curve for this variant is also highly unstable. It reaches 1,123 at iteration 30, falls to 165 at iteration 90, and rebounds to 1,287 at iteration 100. This inconsistency suggests that, although capable of high performance in certain iterations, its erratic behavior limits overall optimization efficiency.

These results confirm that the basic PSO, when calibrated with agricultural-specific parameters, is best suited to optimize sustainable agriculture innovation ecosystems. Its stability aligns with the dynamic nature of rural innovation systems, where resource allocation must balance student skill development, farmer adoption, and ecological enhancement—driven by a commitment to green transformation, incentive mechanisms for sustainable practices, and knowledge diffusion from universities to farms. By mirroring the iterative logic of ecological adaptation, PSO ensures that educational resources consistently advance rural green transformation.

CONCLUSIONS

Compared with the SVM algorithm, PSO significantly improved resource allocation in sustainable agriculture education, enhancing precision agriculture resource utilization efficiency by 4.4% (from 0.085 to 0.129) and regional agricultural resource allocation efficiency by 9.2% (from 0.135 to 0.227). This optimization was achieved by dynamically adjusting the quantity and structure of agricultural-specific resources. In terms of quantity, it optimized the ratio of sustainable agriculture faculty to students—for example, reducing the student-to-instructor ratio for hands-on drone irrigation training from 30:1 to 15:1. Structurally, it allocated 38% more funding to majors aligned with regional farming needs, such as prioritizing Agricultural Resources and Environment programs in Henan's dryland grain regions and Agronomy programs in Jiangsu's cash crop areas. These adjustments reduced evaluation errors (e.g., over-reliance on entrepreneurial project counts) and strengthened the system's capacity to assess both educational and ecological outcomes.

The constructed evaluation system encompasses three agriculture-focused core levels and 28 scientifically validated indicators:

- Agricultural platform support environment: Includes the number of university–cooperative incubators, partnerships with agricultural extension centers, and precision agriculture laboratory equipment.
- Sustainable resource allocation environment: Covers per-student funding for low-carbon farming training and the proportion of resources directed to regional ecological priorities (e.g., water-saving technology in arid areas).
- Sustainability-oriented outcomes: Encompasses student ecological literacy scores, farmer adoption rates of student-developed technologies, and GHG reduction rates on partner farms.

The system integrates qualitative methods (semi-structured interviews with 15 experts, including agricultural extension officers and agritech enterprise managers) and quantitative methods (questionnaire data from 1,200 agricultural majors), avoiding the one-dimensionality of traditional evaluations that overemphasize competition results or project counts while neglecting ecological impact. Furthermore, the basic PSO algorithm demonstrated strong stability in agricultural contexts: the standard deviation of its global minimum value for key optimization functions (e.g., farmer technology adoption, water-use efficiency) was ≤ 0.17352, and the effect of sample size variation (e.g., expansion from 1,200 to 1,500 students) on the model was less than 5%. This performance exceeded that of improved algorithms such as QPSO and PSO-In in convergence consistency, particularly in adapting to variable rural conditions, including seasonal fluctuations in training needs between planting and harvest periods.

To translate these findings into actionable strategies for sustainable agriculture innovation education, targeted policy recommendations can be developed for different agricultural contexts. For resource-poor agricultural regions (e.g., arid Northwest China and grain-producing areas in central Henan), the focus should be on optimizing soft infrastructure. The PSO algorithm can be used to adjust funding ratios among agricultural majors—for example, increasing investment in Agronomy programs specializing in drought-resistant crop cultivation to align with local dryland farming needs—and to develop high-quality online courses (e.g., Precision Soil Moisture Management for Wheat) that compensate for limited physical resources such as offline precision agriculture laboratories.

For economically developed agricultural regions (e.g., Zhejiang's cash crop areas and Shandong's facility agriculture hubs), the priority is to strengthen school–enterprise–farmer collaboration. PSO can be applied to determine the optimal number of cooperative projects (e.g., 12 student–agritech startup partnerships per semester) and on-farm internship placements (e.g., 80 internships annually with organic vegetable cooperatives), ensuring that resources are directed toward high-demand green fields such as smart agriculture (IoT-based crop monitoring) and circular agriculture (agri-waste recycling). This approach bridges the gap between classroom instruction and on-farm practice, enabling students to test and refine innovations in real ecological settings.

For global rural agricultural regions (e.g., Southeast Asian smallholder farming areas and African subsistence agriculture hubs), the evaluation system can be localized by increasing the weight of context-specific indicators such as "alignment with local smallholder needs" (e.g., suitability for low-input maize farming) and "access to micro-irrigation training" (rather than generic entrepreneurship loans). This adaptation reflects the realities of informal agriculture and limited educational infrastructure, enhancing the global applicability of the research and supporting smallholder green transformation.

Despite its contributions, this study has limitations that suggest directions for future research.

- Sample scope constraints: The research sample includes eight Chinese agricultural institutions across three majors—Agricultural Resources and Environment, Agronomy, and Agribusiness Management—potentially limiting generalizability to specialized contexts (e.g., tropical agriculture in Yunnan or animal husbandry in Inner Mongolia). Future studies could test the system in a wider range of institutions, including vocational agricultural colleges and tropical agriculture universities, to evaluate its adaptability across diverse sectors.
- Long-term impact verification: The current study emphasizes short-term outcomes (e.g., post-course skill mastery and one-year startup survival). A five-year longitudinal study is needed to examine the correlation between evaluation results and long-term ecological effects—such as sustained soil fertility improvements from student-designed organic fertilizer solutions or persistent GHG reductions on farms adopting student technologies—to verify the system's predictive validity for lasting rural green development.
- Dynamic optimization potential: The existing system employs static indicators (e.g., annual resource allocation). Future research could integrate real-time agricultural IoT data (e.g., soil moisture and crop yield fluctuations) with the PSO algorithm to develop a dynamic evaluation model. Such a system would update indicators and weights in real time—for instance, increasing the weight of flood-resilient farming training during rainy seasons—enhancing timeliness and accuracy while better supporting the cultivation of sustainable agriculture innovators within the global food security and low-carbon agriculture agenda.

DATA AVAILABILITY

All figures and tables supporting the findings of this study are included within the article.

COMPETING INTERESTS

The authors of this publication declare there are no competing interests.

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AUTHOR'S CONTRIBUTION

Jing Qiao was responsible for the drafting and revision of the manuscript.

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